

# Robert Rosenbaum

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## Professional Experience

### **Associate Professor with Tenure (2020 - Present)**

University of Notre Dame  
Department of Applied and Computational Mathematics and Statistics  
Department of Biological Sciences (concurrent)

### **Director of Graduate Studies (2021 - Present)**

University of Notre Dame  
Department of Applied and Computational Mathematics and Statistics

### **Huisking Foundation Assistant Professor of Applied and Computational Mathematics and Statistics (2019 - 2020)**

University of Notre Dame  
Department of Applied and Computational Mathematics and Statistics

### **Assistant Professor (2014 - 2019)**

University of Notre Dame  
Department of Applied and Computational Mathematics and Statistics

### **Postdoctoral Associate (2011 - 2014)**

University of Pittsburgh  
Department of Mathematics

## Education

### **Ph.D. and M.S. in Mathematics (2011, 2008)**

University of Houston

### **B.S. in Mathematics and in Computer Science (2006)**

University of Houston

## Publications

1. Z. Ding, ... , R. Rosenbaum, ... , and A. Tolia. Functional connectomics reveals general wiring rule in mouse visual cortex. *Nature*, 640(8058):459–469, 2025
2. Y. Wan and R. Rosenbaum. High-dimensional dynamics in low-dimensional networks. *arXiv preprint arXiv:2504.13727*, 2025

3. N. Shervani-Tabar, M. A. Mirhoseini, and R. Rosenbaum. Oja’s plasticity rule overcomes several challenges of training neural networks under biological constraints. *arXiv preprint arXiv:2408.08408*, 2025
4. R. Rosenbaum. *Modeling Neural Circuits Made Simple with Python (Textbook)*. MIT Press, 2024
5. V. Zhu and R. Rosenbaum. Learning fixed points of recurrent neural networks by reparameterizing the network model. *Neural Computation*, 36(8):1568–1600, 2024
6. A. R. Andrei, A. E. Akil, N. Kharas, R. Rosenbaum, K. Josić, and V. Dragoi. Rapid compensatory plasticity revealed by dynamic correlated activity in monkeys in vivo. *Nature Neuroscience*, pages 1–10, 2023
7. N. Shervani-Tabar and R. Rosenbaum. Meta-learning biologically plausible plasticity rules with random feedback pathways. *Nature Communications*, 14(1):1805, 2023
8. R. Rosenbaum. On the relationship between predictive coding and backpropagation. *PLoS One*, 17(3):e0266102, 2022
9. V. Zhu and R. Rosenbaum. Evaluating the extent to which homeostatic plasticity learns to compute prediction errors in unstructured neuronal networks. *Journal of Computational Neuroscience*, pages 1–17, 2022
10. A. E. Akil, R. Rosenbaum, and K. Josić. Balanced networks under spike-time dependent plasticity. *PLOS Computational Biology*, 17(5):e1008958, 2021
11. B. C. Schwab, D. Kase, A. Zimnik, R. Rosenbaum, M. G. Codianni, J. E. Rubin, and R. S. Turner. Neural activity during a simple reaching task in macaques is counter to gating and rebound in basal ganglia–thalamic communication. *PLoS Biology*, 18(10):e3000829, 2020
12. C. Baker, V. Zhu, and R. Rosenbaum. Nonlinear stimulus representations in neural circuits with approximate excitatory-inhibitory balance. *PLoS Computational Biology*, 16(9):e1008192, 2020
13. C. Ebsch and R. Rosenbaum. Spatially extended balanced networks without translationally invariant connectivity. *Journal of Mathematical Neuroscience*, 10(1):1–14, 2020
14. C. Baker, E. Froudarakis, D. Yatsenko, A. Tolias, and R. Rosenbaum. Inference of synaptic connectivity and external variability in neural microcircuits. *Journal of Computational Neuroscience*, pages 1–25, 2020
15. C. Huang, D. Ruff, R. Pyle, R. Rosenbaum, M. Cohen, and B. Doiron. Circuit models of low-dimensional shared variability in cortical networks. *Neuron*, 101(2):337–348, 2019
16. C. Baker, C. Ebsch, I. Lampl, and R. Rosenbaum. Correlated states in balanced neuronal networks. *Physical Review E*, 99(5):052414, 2019
17. R. Pyle and R. Rosenbaum. A reservoir computing model of reward-modulated motor learning and automaticity. *Neural Computation*, 31(7):1430–1461, 2019
18. C. Ebsch and R. Rosenbaum. Imbalanced amplification: A mechanism of amplification and suppression from local imbalance of excitation and inhibition in cortical circuits. *PLoS Computational Biology*, 14(3):e1006048, 2018
19. R. Rosenbaum, M. Smith, A. Kohn, J. Rubin, and B. Doiron. The spatial structure of correlated neuronal variability. *Nature Neuroscience*, 20(1):107, 2017

20. G. Koch Ocker, Y. Hu, M. Buice, B. Doiron, K. Josić, R. Rosenbaum, and E. Shea-Brown. From the statistics of connectivity to the statistics of spike times in neuronal networks. *Current Opinion in Neurobiology*, 46:109–119, 2017
21. R. Pyle and R. Rosenbaum. Spatiotemporal dynamics and reliable computations in recurrent spiking neural networks. *Physical Review Letters*, 118(1):018103, 2017
22. B. Doiron, A. Litwin-Kumar, R. Rosenbaum, G. Ocker, and K. Josić. The mechanics of state-dependent neural correlations. *Nature Neuroscience*, 19(3):383–393, 2016
23. R. Pyle and R. Rosenbaum. Highly connected neurons spike less frequently in balanced networks. *Physical Review E*, 93:040302, 2016
24. R. Rosenbaum. A diffusion approximation and numerical methods for adaptive neuron models with stochastic inputs. *Frontiers in Computational Neuroscience*, 10, 2016
25. A. Litwin-Kumar, R. Rosenbaum, and B. Doiron. Inhibitory stabilization and visual coding in cortical circuits with multiple interneuron subtypes. *Journal of Neurophysiology*, 92:3844–3848, 2016
26. R. Rosenbaum and B. Doiron. Balanced networks of spiking neurons with spatially dependent recurrent connections. *Physical Review X*, 4(2):021039, 2014
27. R. Rosenbaum, T. Tchumatchenko, and R. Moreno-Bote. Correlated neuronal activity and its relationship to coding, dynamics and network architecture. *Frontiers in Computational Neuroscience*, 8, 2014
28. R. Rosenbaum, A. Zimnik, F. Zheng, R. Turner, C. Alzheimer, B. Doiron, and J. Rubin. Axonal and synaptic failure suppress the transfer of firing rate oscillations, synchrony and information during high frequency deep brain stimulation. *Neurobiology of Disease*, 62:86–99, 2014
29. R. Rosenbaum. Short term plasticity, biophysical models. In *Encyclopedia of Computational Neuroscience*, pages 1–5. Springer, 2014
30. S. Reich and R. Rosenbaum. The impact of short term synaptic depression and stochastic vesicle dynamics on neuronal variability. *Journal of Computational Neuroscience*, 35(1), 2013
31. R. Rosenbaum, J. Rubin, and B. Doiron. Short term synaptic depression and stochastic vesicle dynamics reduce and reshape neuronal correlations. *Journal of Neurophysiology*, 108(2), 2013
32. R. Rosenbaum, J. Rubin, and B. Doiron. Short term synaptic depression imposes a frequency dependent filter on synaptic information transfer. *PLoS Computational Biology*, 8(6), 2012
33. A. Hazra, R. Rosenbaum, B. Bodmann, S. Cao, K. Josić, and J. Ziburkus. Beta adrenergic modulation of spontaneous spatiotemporal activity patterns and synchrony in hyper-excitable hippocampal circuits. *Journal of Neurophysiology*, 108(2), 2012
34. R. Rosenbaum and K. Josić. Membrane potential and spike train statistics depend distinctly on input statistics. *Physical Review E*, 84, Nov 2011
35. R. Rosenbaum, J. Trousdale, and K. Josić. The effects of pooling on correlated neural variability. *Frontiers in Neuroscience*, 5(58), 2011
36. R. Rosenbaum and K. Josić. Mechanisms that modulate the transfer of spiking correlations. *Neural Computation*, 23(5), Jan 2011

37. R. Rosenbaum, F. Marpeau, J. Ma, A. Barua, and K. Josić. Finite volume and asymptotic methods for stochastic neuron models with correlated inputs. *Journal of Mathematical Biology*, 35(1), 2011
38. R. Rosenbaum, J. Trousdale, and K. Josić. Pooling and correlated neural activity. *Frontiers in Computational Neuroscience*, 4(9), 2010
39. K. Josić and R. Rosenbaum. Unstable solutions of nonautonomous linear differential equations. *SIAM Review*, 50(3), 2008

## Research Funding

Over \$2 million in funding in total.

### **DoD MURI Award 2025-2030**

Award amount: \$883,001 to Rosenbaum (\$7,500,000 total)

Title: *Multiscale Neural Mechanisms Balancing Cognitive Costs Under Energy Constraints*

Role: Sole PI at Notre Dame. Grant is joint with 6 other PIs at other universities.

### **Air Force DEPSCOR Award 2021-2025**

Award amount: \$600,000

Title: *Using Meta-plasticity to Discover the Biophysics of Learning*

Role: Lead PI. Grant is joint with one other PI at Notre Dame.

### **NSF NeuroNex Theory Team 2017-2023 (DBI-1707400)**

Award amount: \$387,096 to Rosenbaum (\$4,393,191 total)

Title: *Inferring interactions between neurons, stimuli, and behavior*

Role: Sole PI at Notre Dame. Grant is joint with 4 other PIs at other universities and one contracted researcher.

### **NSF CAREER 2017-2022 (DMS-1654268)**

Award amount: \$400,000

Title: *Form and Function in Cortical Neuronal Networks*

Role: Sole PI

### **Huisking Foundation, Inc. Award 2019-2020**

Award amount: \$5,000

### **NSF Mathematical Biology 2015-2018 (DMS-1517828)**

Award amount: \$180,000

Title: *Statistical Dynamics of Balanced Cortical Networks*

Role: Sole PI

## Other Funding

### **NSF Funding for 2017 International Conference on Mathematical Neuroscience (DMS-1642544)**

Award amount: \$20,000.

Role: Co-PI

**Burroughs Wellcome Funding for 2017 International Conference on Mathematical Neuroscience 2016-2017 (1016529)**

Award amount: \$5,000

Role: Co-PI

**SIAM Funding for 2017 International Conference on Mathematical Neuroscience 2016-2017**

Award amount: \$5,100

Role: Co-PI

## **Awards and Honors**

**Edmund P. Joyce Award for Excellence in Undergraduate Teaching 2022**

**Huisking Foundation Assistant Professor Appointment. 2019-2020**

**NSF CAREER Award. 2017-2022**

## **Research Supervision**

### **Postdocs**

Navid Shervani-Tabar (current postdoc)

Marzieh Alireza Mirhoseini (former postdoc)

### **PhD Students**

Jiayin Han (current PhD student)

Matthew Senese (current PhD student)

Vicky Zhu (former PhD student. First position: Tenure track professor at Babson College)

Cody Baker (former PhD student. First position: Data Scientist at CatalystNeuro)

Christopher Ebsch (former PhD student. First position: Researcher at Pacific NW Natl Lab)

Ryan Pyle (former PhD student. First position: Postdoc at Rice University)

### **Undergraduate Students**

Romy Peterson (current undergraduate student)

Yue Wan (former undergraduate student)

Sarah Duessing (former undergraduate student)

James Galante (former undergraduate student)

Maria Pope (former undergraduate student)

David Connelly (former undergraduate student)

Gabrielle Thivierge (former undergraduate student)

Nikhat Dharani (former undergraduate student, co-advised)

Steven Reich (former undergraduate student, co-advised)

## **Pedagogical Activities and Service**

**Director of Graduate Studies in ACMS (2021-Present)**

**ACMS Graduate Studies Committee (2016-Present)**

**Author of undergraduate Computational Neuroscience textbook.**

“Modeling Neural Circuits Made Simple with Python”

Published by MIT Press in 2024

Open access book and code available at:

<https://mitpress.mit.edu/9780262548083/>

<https://github.com/RobertRosenbaum/ModelingNeuralCircuits>

**Courses Designed:**

Artificial Neural Networks (undergraduate)

Mathematical and Computational Modeling in Neuroscience (undergraduate)

Deep Learning (graduate)

Computational Neuroscience (graduate)

Optimization and Deep Neural Networks (graduate)

**Steering committee for the Notre Dame Neuroscience and Behavior undergraduate major (2017-2019)**

## Organizational and Scholarly Activities

**Grant Proposal Reviewing**

Reviewer for European Research Council Advanced Grant (2026)

Served on NSF panels (2022, 2022, 2023).

Ad Hoc Reviewer for DoD Army Basic Scientific Research (2021)

**Computational and Systems Neuroscience (COSYNE)**

Program Committee (2018, 2019, 2020)

Session Chair (2018)

Abstract Reviewer (2017-2020)

**International Conference on Mathematical Neuroscience (ICMNS) 2017**

Co-organizer (with Z Kilpatrick and J Gjorgjieva)

**Special issue of *Frontiers in Computational Neuroscience: Correlated neuronal activity and its relationship to coding, dynamics and network architecture* (2013)**

Guest editor (with Tatjana Tchumatchenko and Ruben Moreno-Bote)

**Reviewer for:**

Nature Neuroscience, Nature Communications, Proceedings of the National Academy of Sciences (PNAS) USA, Physical Review Letters, eLife, The Journal of Neuroscience, IEEE Transactions on Neural Networks and Learning Systems, The Journal of Neurophysiology, The Journal of Mathematical Neuroscience, PLoS Computational Biology, PLoS One, Physical Review E, Physical Review X, The Journal of Computational Neuroscience, Frontiers in Computational Neuroscience, Biological Cybernetics

**Guest Editor for:**

PLoS Computational Biology, Frontiers in Computational Neuroscience

**Workshop on Linking Neuronal Network Architecture and Collective Dynamics at SIAM Conference on Applications of Dynamical Systems (2011)**

Co-organizer (with Ashok Kumar)

**Texas Applied Mathematics Meeting for Students (2010)**

Co-organizer (with Chinmaya Gupta)

## **Outreach Activities**

**Collaboration with Riley High School Robotics Team (2017-2019)**

**Introduction to scientific computing for Riley High School students  
(2015-2018)**